

ZHE JIA, Ph.D.

AI Researcher / Principal Investigator • AI4Science, LLMs & Foundation Models

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🐙 github.com/jiazhe868 🔗 [linkedin.com/in/zhe-jia-phd](https://www.linkedin.com/in/zhe-jia-phd) 📄 Google Scholar (20+ Papers)

Professional Summary

Machine Learning Researcher and Principal Investigator building Foundation Models for complex systems, PDE dynamics, and LLM pretraining. Caltech Ph.D. with 8+ years bridging computational physics, distributed HPC, and deep learning. Demonstrated expertise across the **full LLM lifecycle** (Pretraining, SFT, GRPO/RLHF), **mechanistic interpretability**, and **out-of-distribution generalization**. Author of 20+ peer-reviewed papers, including two **Science** papers. Combines framework-level pytorch implementation, distributed computing, and controlled evaluation with research leadership.

Technical Skills

ML Architectures:	LLMs, Graph/Set Transformers, Fourier Neural Operators (FNO), State Space Models, Latent Diffusion.
Pretrain & Align:	PyTorch, Attention mechanisms, FlashAttention, DDP, GRPO / RLVR, SFT.
Physics & Data Sci:	Fluid/Solid Dynamics, PDE Solvers, Out-of-Distribution (OOD) Analysis, RAG, Spectral Methods.
Languages & Tools:	Python, C++, Fortran, Bash, Modular Software Design, Linux, HPC Slurm, Git.

Selected ML Systems & Foundation Models

nanoGPT-Seis & Post-Training: The Full LLM Lifecycle

[Pretrain Repo](#) | [Post-train Repo](#)

- **Pretraining & Scaling:** Engineered a 113M-parameter LLM from scratch. Implemented custom BPE tokenization, MinHash/LSH data deduplication, and modern transformer architectures (GQA, RoPE, FlashAttention-2). Executed IsoFLOP scaling sweeps using Maximal Update Parametrization (muP) on multi-GPU DDP clusters.
- **Alignment (SFT & GRPO):** Transitioned the base model to an instruction-following assistant via Supervised Fine-Tuning. Implemented Group Relative Policy Optimization (GRPO)-style from scratch with programmatic verifiers, raising reasoning (<think>) format compliance from 73% to **98%**. Audited reward exploitation and evaluation leakage, separating format compliance gains from statistically unresolved closed-book numerical accuracy changes.
- **Evaluation & RAG Grounding:** Systematically diagnosed reward hacking behaviors (e.g., tag inflation, denominator dilution). Integrated open-book RAG retrieval to clear closed-book arithmetic bottlenecks, enabling exact multi-step scientific derivations.

Sensoformer: Sim-to-Real for Irregular Physical Sensor Sets

[NeurIPS 2026 under review](#)

- **Architecture Design:** Built a variable-cardinality SciML framework utilizing Siamese 1D-ResNet towers and masked Set-Transformers. Achieved constant-depth global information exchange without iterative graph message-passing.
- **Sim-to-Real Alignment:** Designed *Physics-Structured Domain Randomization* to bridge the reality gap, delivering **1000x lower-latency** inference versus traditional scientific inversions with zero real-world labels.

GTS: Graph Transformer Simulator for Ultra-Long Earth Dynamics

[ICLR in prep](#)

- **Continuous State Space Modeling:** Architected an autoregressive simulator capable of **10,000+ step** rollouts across physical dynamics spanning 15 orders of magnitude. Addressed compounding drift and memory-wall bottlenecks common in AI weather models.
- **Hardware Acceleration:** Leveraged PyTorch 2.0 compiler and CUDA Graph Capture with adaptive time-stepping to eliminate CPU overhead, achieving a **3x zero-loss forward GPU acceleration**.

Mechanistic Interpretability & OOD Generalization in Neural Operators

[Under review](#)

- **Causal Probing & Blind Spots:** Developed a causal Fourier-Jacobian probe to audit off-manifold mechanisms in neural operators. Proved that standard training leaves unexcited directions unidentified, exposing a critical **28% out-of-distribution (OOD) error** under simple input offsets.
- **Symmetry & Structural Repair:** Discovered that standard data canonicalization destroys PDE symmetries. Formulated an exact structural completion that repairs these blind spots without extra data, bridging deep learning interpretability with PDE theory.

Attention Capacity & Information Bottlenecks in Graph Transformers

[Under review](#)

- **Theoretical Bounds:** Formalized "receiver-query capacity" to evaluate how many distinct global summaries Attention can deliver versus Message Passing. Established a random-access lower bound mapping query modes to exact-recovery thresholds.
- **Predictive Capacity Scaling:** Engineered a causal intervention by clustering receiver queries without altering parameter count. Built a frozen predictor that successfully forecasted the exact **95% capacity saturation point** on unseen datasets (PascalVOC-SP).

Professional Experience

Assistant Research Professor / Principal Investigator
The University of Texas at Austin

2024 – Present
Austin, TX

- Lead an AI4Science lab translating complex physical inverse problems into scalable ML frameworks: Set Transformer Models for irregular sensor sets (Sensoformer), graph neural simulators, and uncertainty-aware generative physical AI.
- Own end-to-end model architecture, software design, and compute planning. Manage roadmap and mentor Ph.D./postdoc researchers across ML, HPC, and physics workflows.
- Secured **5M+ CPU-hours/10k+ GPU-hours** in supercomputing allocations and engineered **10TB+** automated data pipelines for unstructured mesh and sensor datasets.

Cecil H. and Ida M. Green Fellow / Technical Lead

University of California, San Diego

2022 – 2024

San Diego, CA

- Led a cross-institutional team combining nonlinear dynamic rupture simulation, large-scale geophysical inversion, and distributed HPC (OpenMPI) to resolve the complex 2023 Kahramanmaraş earthquake doublet (**Science Cover**).
- Engineered robust ETL systems for multi-TB, extreme low-SNR time-series data to identify cascade mechanisms.

Graduate Researcher (Scientific Computing & ML)

California Institute of Technology (Caltech)

2016 – 2022

Pasadena, CA

- Developed a continuous CNN-UNet deep learning framework for signal extraction, delivering a **100x speedup** over classical Digital Signal Processing (DSP) workflows and enabling scalable global data analysis.
- Built highly parallelized Bayesian inversion engines using hybrid MCMC and adjoint optimization for high-dimensional uncertainty quantification.

Education

Ph.D. in Computational Geophysics (Inverse Problems & Optimization)

California Institute of Technology (Caltech)

2022

Pasadena, CA

B.S. & M.S. (Special Class for the Gifted Young - Top 1%)

University of Science and Tech. of China (USTC)

2013 / 2016

Hefei, China

Selected Publications

Selected from 23 peer-reviewed papers (over 1,200 citations) spanning ML and Science venues. (: Lead or advisory author)*

- Jia, Z.*, (2026). *Mechanisms Off the Data Manifold: Distribution-Relative Identifiability in Neural Operators*. **Under review**.
- Jia, Z.*, et al. (2026). *How Many Global Summaries Does Attention Need? Receiver-Query Capacity in Graph Transformers*. **Under review**.
- Jia, Z.*, Zhang, X., Li, J. (2026). *Sensoformer: Robust Sim-to-Real Inference on Variable-Geometry Sensor Sets*. **NeurIPS** (in review). [\[pdf\]](#)
- Liu, D., Jia, Z.*, et al. (2026). *GTS: A Graph Transformer Foundation Model for Ultra-Long Autoregressive Dynamics*. **ICLR** (in prep).
- Zhang, X., Jia, Z.*, Ross, Z., Clayton, R. (2020) *Extracting dispersion curves from ambient noise correlations using deep learning*. **IEEE TGRS**.
- Jia, Z.*, et al. (2023) *The complex dynamics of the 2023 Kahramanmaraş, Turkey, Mw 7.8-7.7 earthquake doublet*. **Science** (Cover). - Large-scale HPC multi-modal inversion (200+ citations).
- Ross, Z., Idini, B., Jia, Z., et al. (2019) *Hierarchical interlocked orthogonal faulting in the 2019 Ridgecrest earthquake sequence*. **Science**. - High-impact dynamic system analysis (500+ citations).

Selected Honors & Awards

National Science Foundation's Grant Award | \$440k award for research

2026

Cecil H. and Ida M. Green Fellowship | \$138k independent award for research excellence

2022 – 2024

Einstein Visa for Extraordinary Ability

2023

Outstanding Self-Financed Student Award | National Award, Top 1%

2023

Caltech GPS Fellowship | Full graduate study fellowship

2016

China National Scholarship | Top 1% of graduate students nationwide

2015